

Micro 4 Final Question 3

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Consider the seller of a single indivisible object who values it at 0, together with a buyer whose outside option is 0. The buyer's valuation of the object is v_h with probability π and v_l with probability $1 - \pi$; not consuming the object is worth 0 to the buyer. As before, the buyer and seller can trade money, and the seller can design a selling mechanism as a take-it-or-leave-it offer.

1. What is the seller's optimal mechanism? How does it depend on π ?

The seller's optimal mechanism is to offer a price p that is either v_l or v_h based on the quantity of high types π . The indifference point is defined by

$$v_l = \pi^* v_h \implies \pi^* = \frac{v_l}{v_h}.$$

Thus, the seller sets p such that

$$p = \begin{cases} v_l & \text{if } \pi < \pi^* \\ v_h & \text{if } \pi \geq \pi^*. \end{cases}$$

2. The buyer's "prior" utility is his expected payoff from the optimal mechanism before learning his type. How does the buyer's prior utility depend on π ?

If $\pi < \pi^*$, then $p = v_l$. All types buy the good, but only the high types receive a surplus equal to $v_h - v_l$. When $\pi \geq \pi^*$, $p = v_h$. Only the high types buy the good, but they receive no surplus. Thus, the expected utility EU is

$$EU = \begin{cases} \pi(v_h - v_l) & \text{if } \pi < \pi^* \\ 0 & \text{if } \pi \geq \pi^*. \end{cases}$$

3. Suppose that the buyer can send to the seller a partially informative signal about his valuation, subject to Bayes' rule. What is the maximal prior utility that a buyer could obtain with the help of such a signal as a function of π ? What posterior beliefs would the buyer induce on the seller in this case? What about deadweight loss?

First, note that if $\pi < \pi^*$, the buyer is not incentivized to send any signal: the high types are already buying the good at the low price v_l . So assume that $\pi \geq \pi^*$. The buyer can send a signal $s \in \{h, l\}$. He commits to a signaling structure $P(h|v_h), P(l|v_h), P(h|v_l), P(l|v_l)$. For example, $P(h|v_h)$ is the probability that the buyer will send a "high" signal given that his valuation is v_h . First, note that low types will not want to signal that they are a high type. Thus, $P(h|v_l) = 0$ and $P(l|v_h) = 1$. However, the high types want to trick the seller into selling them the good at price v_l . They can't always lie, or else the seller will revert to charging the high price v_h . Instead, they lie just enough so that, given the seller sees signal l , he is indifferent between charging v_l and v_h :

$$v_l = P(v_h|l)v_h.$$

If the seller charges v_l , he is guaranteed to sell the good. If he charges v_h , he will only sell the good if the buyer is lying as a high type. Thus, the buyer must induce the following posterior belief:

$$P(v_h|l) = \frac{v_l}{v_h}.$$

Also, by Bayes' rule, we know that

$$P(v_h|l) = \frac{P(l|v_h)P(v_h)}{P(l)}.$$

Therefore, the buyer must commit to $P(l|v_h)$ satisfying

$$\frac{v_l}{v_h} = \frac{P(l|v_h)\pi}{(1-\pi) + P(l|v_h)\pi}.$$

Solving for $P(l|v_h)$ yields

$$P(l|v_h) = \frac{v_l(1-\pi)}{\pi(v_h - v_l)}.$$

We can now derive the buyer's expected utility given this signaling structure: When $\pi \geq \pi^*$, the buyer receives a surplus if he is the high type that lies about

his type and is charged a low price: $\pi P(l|v_h)(v_h - v_l) = v_l(1 - \pi)$. Since the buyer does not send an informative signal if $\pi < \pi^*$, the new expected utility given π is

$$EU = \begin{cases} \pi(v_h - v_l) & \text{if } \pi < \pi^* \\ v_l(1 - \pi) & \text{if } \pi \geq \pi^*. \end{cases}$$

The posterior beliefs that this signal structure induces are

$$P(v_h|h) = 1$$

$$P(v_h|l) = \frac{v_l}{v_h}$$

$$P(v_l|h) = 0$$

$$P(v_l|l) = 1 - \frac{v_l}{v_h}.$$

(Note that we derived $P(v_h|l)$ earlier; the others can be derived via Bayes' rule). There is no longer any deadweight loss: all types of buyers are able to buy the good now.